

CHE 201 Spring 2020 HW 4

Roxanna Mendoza

TOTAL POINTS

94.5 / 100

QUESTION 1

1 Question 1 20 / 20

✓ - **0 pts** Correct

- **5 pts** "...in terms of their individual thermal efficiencies"

✓ - **2.5 pts** Insufficient component about the phase-out schedule

- **5 pts** Insufficient component about the replacer

- **2.5 pts** Insufficient component about the replacer

- **1 pts** No enough references

QUESTION 2

2 Question 2 20 / 20

✓ - **0 pts** Correct

- **5 pts** Wrong V2

- **5 pts** Wrong work

- **5 pts** Wrong internal energy difference

- **5 pts** Wrong final temperature

- **20 pts** Did not answer 2.53

QUESTION 3

3 Question 3 20 / 20

✓ - **0 pts** Correct

QUESTION 4

4 Question 4 17 / 20

- **0 pts** Correct

- **20 pts** Did not answer problem 3.7

✓ - **3 pts** (a) Wrong sketch of T-v diagram

- **3 pts** (a) Wrong specific volume

- **3 pts** (b) Wrong sketch of T-v diagram

- **3 pts** (b) Wrong specific volume

- **3 pts** (c) Wrong sketch of T-v diagram

- **3 pts** (c) Wrong specific volume

QUESTION 5

5 Question 5 17.5 / 20

- **0 pts** Correct

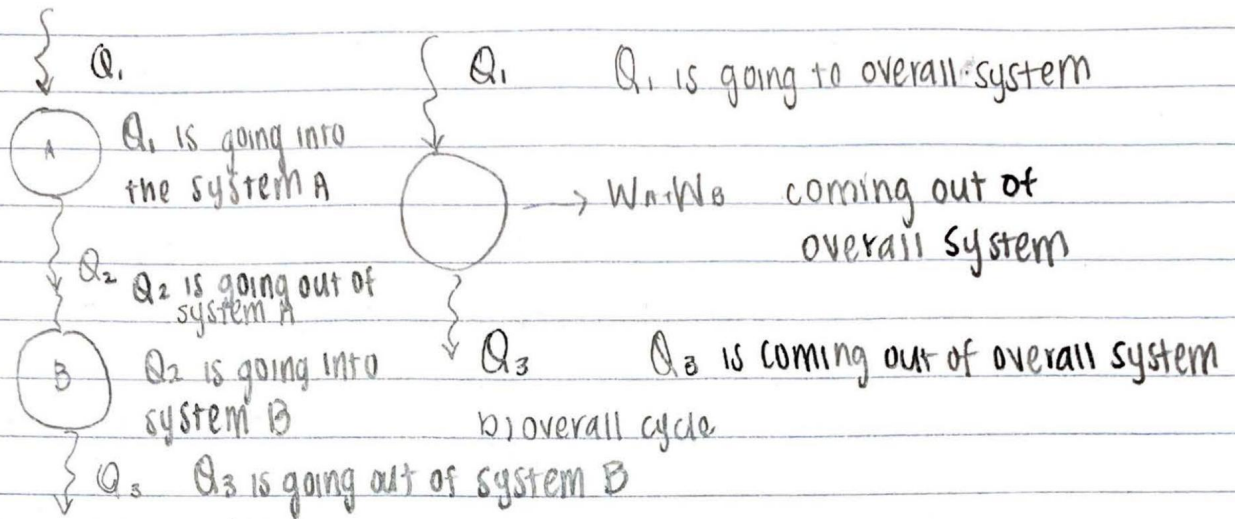
- **20 pts** Did not answer 3.4D

- **5 pts** Insufficient component about the phase-out schedule

CHE 201 homework #4

Roxanna Mendoza

#1.



a) A and B in series

For A

$$\eta_A = \frac{Q_1 - Q_2}{Q_1}$$

$$\frac{Q_2}{Q_1} = 1 - \eta_A$$

$$\dot{W}_A = \dot{Q}_1 \eta_A$$

For B

$$\eta_B = \frac{Q_2 - Q_3}{Q_2}$$

$$\frac{Q_3}{Q_2} = 1 - \eta_B$$

$$\dot{W}_B = \dot{Q}_2 \eta_B$$

For Overall cycle

$$\eta_T = \frac{Q_1 - Q_3}{Q_1} \quad \dot{W}_T = \frac{W_A + W_B}{Q_1}$$

$$W_A = Q_1 \cdot \eta_A$$

$$W_B = Q_2 \cdot \eta_B$$

$$\eta_T = \frac{(Q_1 \cdot \eta_A) + (Q_2 \cdot \eta_B)}{Q_1}$$

$$\eta_T = \frac{Q_1 (\eta_A)}{Q_1} + \frac{Q_2 (\eta_B)}{Q_1} \Rightarrow \eta_T = \eta_A + \frac{Q_2 \eta_B}{Q_1}$$

we know $\frac{Q_2}{Q_1} = 1 - \eta_A$ $\therefore$

$$\eta_T = \eta_A + (1 - \eta_A) \eta_B$$

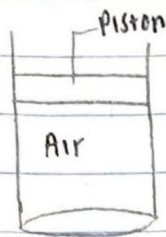
$$\eta_T = \eta_A + \eta_B - \eta_A \eta_B$$

1 Question 1 20 / 20

✓ - 0 pts Correct

- 5 pts "...in terms of their individual thermal efficiencies"

#2.



• $p_1 = 100 \text{ lbf/in}^2$; $v_1 = 3.704 \text{ ft}^3/\text{lb}$; $T_1 = 1000^\circ\text{R}$

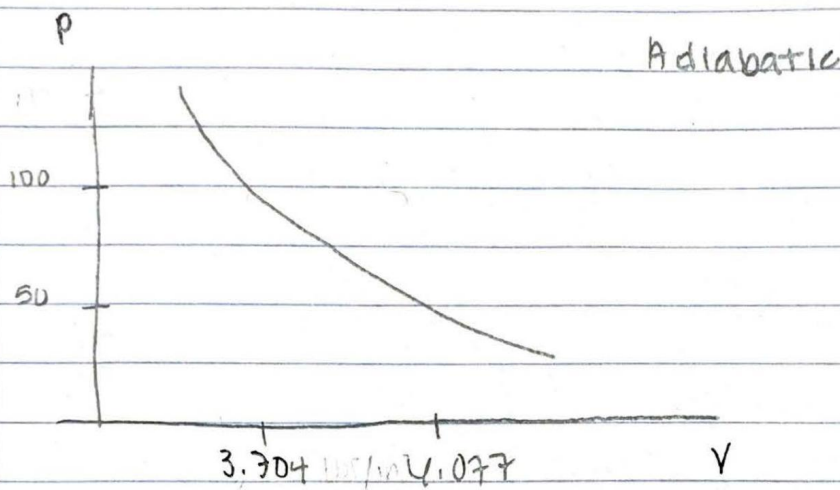
• $P_2 = 50 \text{ lbf/in}^2$

• Polytropic $PV^n = C$ with $n = 1.4$



• $\Delta U = (0.171)(T_2 - T_1)$

$P = \text{Constant}/V^n$



$W = \int_{v_1}^{v_2} P dv$

$W = \int_{v_1}^{v_2} \frac{C}{v^{1.4}} dv = \frac{P_2 v_2 - P_1 v_1}{1 - 1.4} \quad \therefore v_2 = \left(\frac{P_1}{P_2} \right)^{1/1.4} v_1$

$\left(\frac{100 \text{ lbf/in}^2}{50 \text{ lbf/in}^2} \right)^{1/1.4} (3.704 \text{ ft}^3/\text{lb}) = 6.077 \text{ ft}^3/\text{lb}$

$W = \frac{(50 \text{ lbf/in}^2)(6.077 \text{ ft}^3/\text{lb}) - (100 \text{ lbf/in}^2)(3.704 \text{ ft}^3/\text{lb})}{1 - 1.4}$

$W = 164.375 \left(\frac{144 \text{ in}^2}{1 \text{ ft}^2} \right) \left(\frac{1 \text{ Btu}}{778 \text{ ft} \cdot \text{lbf}} \right) = 30.794 \text{ Btu/lb}$

$$Q = \Delta U + W$$

$$Q = 0 \text{ so } \Delta U = -W$$

$$\Delta U = (0.171)(T_2 - T_1)$$

$$(0.171)(T_2 - T_1) = -30.794 \text{ Btu/lb} \quad T_1 = 1000$$

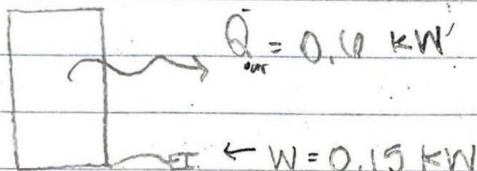
$$(0.171 T_2) - (1000 \times 0.171) = -30.794 \text{ Btu/lb}$$

$$0.171 T_2 - 171 = -30.794$$

$$+171 \quad +171$$

$$T_2 = 819.92^\circ \text{R}$$

#3.



Refrigeration

$$W_{\text{cycle}} = Q_{\text{out}} - Q_{\text{in}}$$

$$\beta = \frac{Q_{\text{in}}}{W_{\text{cycle}}}$$

$$0.15 \text{ kW} = 0.6 \text{ kW} - Q_{\text{in}}$$

$$Q_{\text{in}} = 0.45 \text{ kW}$$

$$\beta = \frac{0.45 \text{ kW}}{0.15 \text{ kW}}$$

$$\beta = 3$$

2 Question 2 20 / 20

✓ - 0 pts Correct

- 5 pts Wrong V_2
- 5 pts Wrong work
- 5 pts Wrong internal energy difference
- 5 pts Wrong final temperature
- 20 pts Did not answer 2.53

$$Q = \Delta U + W$$

$$Q = 0 \text{ so } \Delta U = -W$$

$$\Delta U = (0.171)(T_2 - T_1)$$

$$(0.171)(T_2 - T_1) = -30.794 \text{ Btu/lb} \quad T_1 = 1000$$

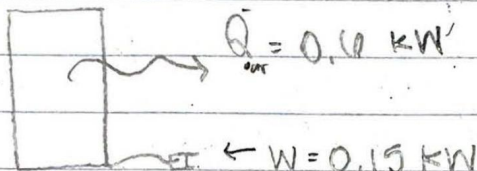
$$(0.171 T_2) - (1000 \times 0.171) = -30.794 \text{ Btu/lb}$$

$$0.171 T_2 - 171 = -30.794$$

$$+171 \quad +171$$

$$T_2 = 819.92^\circ \text{R}$$

#3.



Refrigeration

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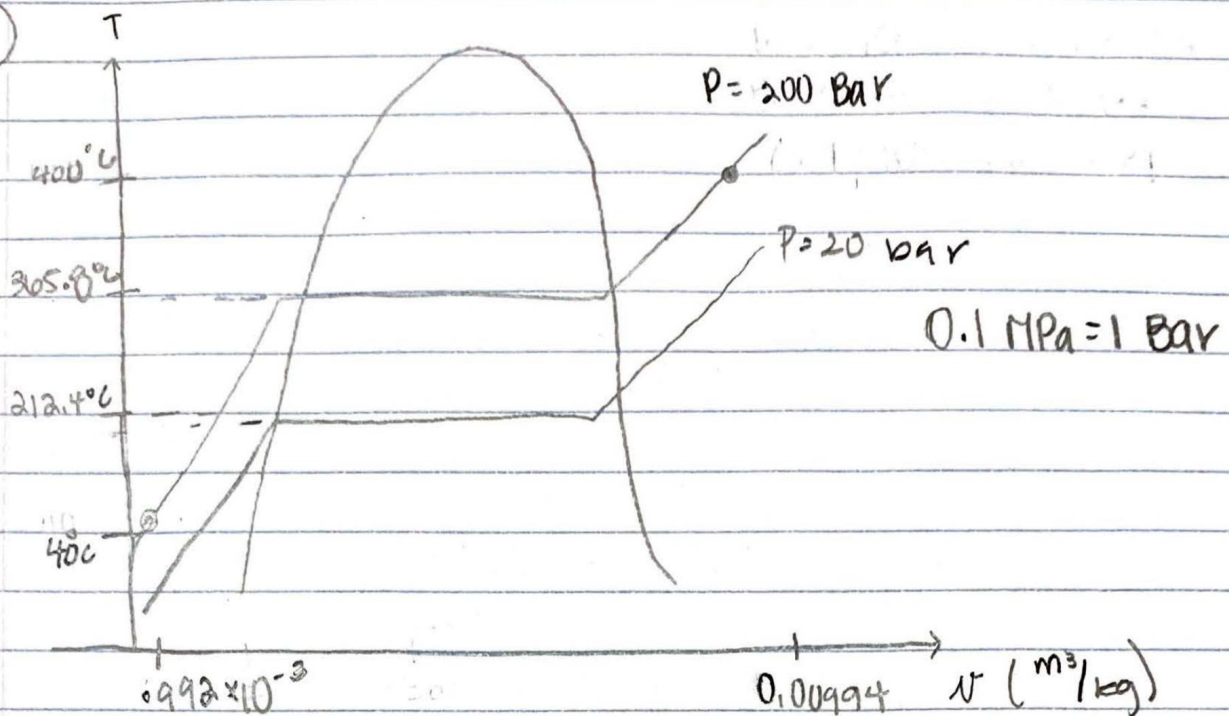
$$\beta = \frac{0.45 \text{ kW}}{0.15 \text{ kW}}$$

$$\beta = 3$$

3 Question 3 20 / 20

✓ - 0 pts Correct

#4.



$$A. \frac{20 \text{ MPa}}{0.1 \text{ MPa}} \bigg| \frac{1 \text{ Bar}}{0.1 \text{ MPa}} = 200 \text{ Bar} \quad \& \quad T = 400^\circ\text{C}$$

$$v_1 = 0.00994 \text{ m}^3/\text{kg} \quad \leftarrow \text{superheated water vapor}$$

$$B. \frac{20 \text{ MPa}}{0.1 \text{ MPa}} \bigg| \frac{1 \text{ Bar}}{0.1 \text{ MPa}} = 200 \text{ Bar} \quad \& \quad T = 40^\circ\text{C}$$

$$v = 0.00992 \times 10^{-3} \text{ m}^3/\text{kg} \quad \leftarrow \text{compressed liquid}$$

$$C. \frac{2 \text{ MPa}}{0.1 \text{ MPa}} \bigg| \frac{1 \text{ Bar}}{0.1 \text{ MPa}} = 20 \text{ Bar} \quad \& \quad T = 40^\circ\text{C}$$

$$v = v_{2.5 \text{ MPa}} + (v_{5 \text{ MPa}} - v_{2.5 \text{ MPa}}) \left(\frac{P - P_{2.5 \text{ MPa}}}{P_{5 \text{ MPa}} - P_{2.5 \text{ MPa}}} \right)$$

$$v = (1.0067 \times 10^{-3}) + (1.0056 \times 10^{-3} - 1.0067 \times 10^{-3}) \left(\frac{2 - 2.5}{5 - 2.5} \right)$$

$$= (1.0067 \times 10^{-3}) + (-0.011 \times 10^{-3}) (-0.2)$$

$$v = 1.0069 \times 10^{-3} \text{ m}^3/\text{kg} \quad \leftarrow \text{compressed liquid}$$

4 Question 4 17 / 20

- 0 pts Correct
- 20 pts Did not answer problem 3.7
- ✓ - 3 pts (a) Wrong sketch of T-v diagram
 - 3 pts (a) Wrong specific volume
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Roxanna Mendoza

16 February 2020

CHE 201

HW 4 Problem 3.4D

R22 has been scheduled for phase out starting 2020 and ending 2029. The Environmental Protection Agency (EPA) is leading for a complete phaseout for all hydrochlorofluorocarbons (HCFCs) in 2030. Although R22 has been highly used for “refrigeration facilities, conditioners and heat pumps”, its harm to the environment brought its downfall.

Current alternatives to R22 are more costly, but often more efficient than its predecessor. R410a, also known as Puron, is a refrigerant that “provides a higher safety rating and an ozone depletion rating of zero, and it performed slightly better on energy-efficiency tests than R22”. A change to R410a is a costly alternative as it would require changing the entire refrigeration system.

Another promising candidate is R407C. This refrigerant shares a lot of the thermodynamic properties as R22. This makes it much more compatible than R410a as it can be “retrofitted in limited Freon systems as long as the process involves a change in oil”.

Works Cited

- “Four Types Of Refrigerants That Are Safer Than Freon.” Do It Yourself, 11 Mar. 2019,
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- “Phaseout of Ozone-Depleting Substances (ODS).” *EPA*, Environmental Protection Agency, 16 Jan. 2020, www.epa.gov/ods-phaseout.
- Service Experts Heating & Air Conditioning. “R22 Refrigerant Is Phasing Out - Updated 2018: Service Experts.” *Service Experts Heating & Air Conditioning*,
www.serviceexperts.com/blog/what-will-happen-to-r22-and-how-does-it-affect-you.
- Studio, CodEX World. “History of Creation.” *AllChem Company* |*| *History of Creation*,
www.allchemi.com/eng/refregerants/history.html.

5 Question 5 17.5 / 20

- **0 pts** Correct
- **20 pts** Did not answer 3.4D
- **5 pts** Insufficient component about the phase-out schedule
- ✓ - **2.5 pts** Insufficient component about the phase-out schedule
- **5 pts** Insufficient component about the replacer
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- **1 pts** No enough references